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I am a Mechanical Engineer with solid hands-on experience spanning upstream oil & gas operations, large-scale EPC projects, and freelance engineering across the oil & gas, petrochemical, and power generation sectors. Currently serving as Environment Engineer (Emission & Energy) at Pertamina Hulu Mahakam, I specialize in GHG emission quantification, energy management, and air pollution control, particularly in **Pengendalian Pencemaran Udara dari Emisi (PPPU)**, covering end-to-end activities from emission monitoring planning, field data collection and review, to regulatory reporting for PROPER compliance.

I hold three BNSP certified professional credentials: **Certified Energy Manager (ISO 50001:2018)**, **Certified Emission Pollution Manager (PPPU)**, and **AK3U (General Occupational Safety and Health)**, reflecting a strong foundation in both technical and regulatory aspects of HSE and sustainability in industrial operations.

Beyond my environmental role, I bring a versatile engineering background that makes me well-suited for roles in **Production Maintenance, Project Engineering, or Project Control**. My experience includes project control and scheduling on one of Indonesia's largest EPCC projects (Freeport Manyar Smelter), alongside freelance mechanical and process engineering work covering ***ASME pressure vessel calculations, CFD/FEM analysis, heat exchanger sizing,*** and process planning for reputable EPC and operator companies. I am open to new opportunities where my mechanical engineering foundation and multi-disciplinary experience can add value.

PROFESSIONAL EXPERIENCE

PT Pertamina Hulu Mahakam (PHM) – Balikpapan Base Office, Kaltim, *Jun 2024 – Present*
Environment Engineer *(Emission & Energy) (TKJP)*

I hold three active BNSP professional certifications; **Certified Energy Manager (ISO 50001:2018), Certified Emission & Air Pollution Manager (PPPU), and AK3U (General Occupational Safety and Health),** supporting energy management, air emission control, decarbonization programs, and GHG monitoring across PHM's upstream oil & gas operations in compliance with API Compendium, IPCC guidelines, and regulatory requirements from SKK Migas and ESDM.

Air Emission Control / PPPU (Pengendalian Pencemaran Udara dari Emisi):

- **Identify** and assess air emission sources across operational facilities, covering combustion equipment, flaring systems, process venting, and fugitive emission sources.
- **Determine** characteristics of emission sources and evaluate emission levels against applicable regulatory thresholds and PROPER compliance criteria (PROPER Biru).
- **Develop and maintain** Emission Monitoring Plans (Rencana Pemantauan Pengendalian Pencemaran Udara) in line with RKL/RPL obligations and SKK Migas requirements.
- **Execute and coordinate** periodic emission monitoring activities, including stack emission testing, data review, and performance evaluation of air pollution control equipment.
- **Compile and submit** emission monitoring results and supporting documents for PROPER Biru compliance reporting to relevant authorities.
- **Maintain operational** documents including Laporan Hasil Uji Udara, Program Pengendalian Pencemaran Udara, Dokumen RKL/RPL, and other

Energy Management & Decarbonization:

- Support decarbonization initiatives covering energy efficiency, emission reduction, and sustainability
- Assist in **Energy Management System** implementation aligned with **ISO 50001:2018**, including data gathering, performance analysis, and standard documentation development.
- Support **Energy Audit** activities including data provision, report review, and **Energy Monitoring System** (ENMS) tools.
- Assist in **ESG and KPI monitoring** within the Sustainability framework.
- Support **Life Cycle Assessment** (LCA) studies requiring energy and emission data.

Energy Consumption & Emission Analysis:

- Calculate **Scope 1 & Scope 2 GHG emissions** from combustion, flaring, process venting, fugitive sources, and electricity consumption (PLN grid).
- Analyze energy consumption data from multiple sources across operations to support efficiency improvement initiatives.
- Monitor and review updates to API Compendium, IPCC guidelines, and Pertamina's emission calculation tools.

Reporting & Documentation:

- Prepare monthly and semester regulatory reports to **SKK Migas, PHE**, and **ESDM**.
- Develop and maintain standard documents related to energy management and emission compliance.

PT Wijaya Karya (Persero) Tbk. (EPCC Division) **– Manyar Smelter Project,** *Jan 2023 – Jun 2024*
Project Control Staff *(Pipe Welding & Inspection Control System Coordinator)* *(Direct Contract)*
Assigned to one of **Indonesia's largest Smelter EPCC projects** under Freeport Indonesia, I was responsible for end-to-end progress monitoring, field coordination, and documentation control for piping works, covering fabrication, inspection, erection, and pre-commissioning phases across a multi-discipline smelter facility.

- Managed progress monitoring and reporting for piping fabrication and erection using **PIC, PST, and RFID-JOVIX systems**, tracking 200,000+ Dia-Inch, 17,000 spools, 346,000 Inch-M across 6 material types, and pipe support fabrication & installation exceeding 400 Ton and 30,000 total quantities.
- Controlled **red-line mark-up** and Site Instructions for piping isometrics, and managed data line lists for Pre-Commissioning and Commissioning test package preparation.
- Coordinated directly with Field Engineers on design changes, supporting **S-Curve achievement** through schedule monitoring, progress tracing, and cross-functional alignment.
- Handled document control and system close-out activities, ensuring all piping records, inspection reports, and completion packages were accurately maintained and traceable for handover.

FREELANCE PROJECT EXPERIENCE

Provided engineering support for multiple oil & gas, petrochemical, and power generation projects in various roles including Mechanical Engineer, Process Engineer, FEM/CFD Specialist, and Project Planner. Responsibilities included ASME compliant mechanical calculations, FEM and CFD analyses, equipment sizing, process planning, and project scheduling for critical packages, ensuring technical compliance, accuracy, and timely execution.

Client & Scope of Work:

- **PT Amman Nusantara Gas** – Sumbawa LNG Terminal & Regas Facility (Nov 2025)
  CFD analysis to evaluate hydrodynamic forces causing accelerated mechanical failure on the CMS.
- **PT Pertamina Hulu Energi ONWJ** – Reactivation of OPL LES Platform (Mar 2025)
  Project Planning for Chemical Injection Package & Slop Pump.
- **Petronas Carigali North Madura II Ltd**. – FEED Hidayah Development Project (Jan 2025)
  Mechanical calculation for Degasser & Electrostatic Coalescer skid packages.
- **PT Tripatra Engineers and Construction** – FEED Bio Refinery Project (Oct 2024)
  ASME pressure vessel calculation for Iron Sponge Absorber (100-V-116 A/B).
- **PT Amman Nusantara Gas** – Sumbawa LNG Terminal & Regas Facility (Apr 2024)
  FEM analysis for pressure vessel stress, displacement, and cycle life.
- **PT Pertamina Hulu Energi** – Upgrading Facility SP AKBG-Stage 1 (Feb 2024)
  Heat exchanger sizing & process planning (shell & tube, plate & frame).
- **PT Aneka Tambang Tbk**. – 2x30 MW CFPP Pomalaa (Aug – Dec 2023)
  CFD simulations for heat transfer & forces in heat exchanger, integrated with FEA stress.
- **PT Medco Energi Internasional Tbk**. – Forel FPSO Time Charter (Nov – Dec 2022)

CFD analysis of hydrodynamic forces on FPSO and liquid levels in LP & HP KO Drum Flare.

EDUCATION EXPERIENCE

UNIVERSITAS NEGERI SURABAYA – Surabaya, Indonesia *2018 – 2022*

Bachelor of Mechanical Engineering; Cumulative GPA: 3.61/4.0

The concentration of Skills: ***Energy*** *(Heat & Mass Transfer)***.**

Final Project Topic: *Studi Eksperimental Pengaruh Sudut Kemiringan Semi-Circular Perforated Baffle dengan Semi-Circular Fins Pada Sisi Annulus Terhadap Performa Double Pipe* ***Heat Exchanger***

TRAINING EXPERIENCE

- Piping Stress Analysis using CAESAR II *May 2023*
- Static Equipment Design using PV Elite *July 2023*
- Intepretation Drawing P&ID, Isometric, Layout, Rotating and Static Equipment *Feb 2023*
- Process Simulation for Basic (Hands-on Simulator Software with Study Case) *Oct – Nov 2022*
- PDMS Piping Mechanical Oil & Gas Design Course *Oct 2022*
- Agile Innovation Project-Based Learning *Aug 2021 – Feb 2022*
- Basic Piping and Instrument Diagram with AutoCAD Application *Sep 2022*
- Project Planning & Controlling Using Ms. Project (Oil and Gas Case) *Sep 2022*
- Introduction Rigging and Lifting with Simulation AutoCAD and Inventor *Aug – Sep 2022*

SKILL

Technical: Microsoft Office (Word, Excel, PowerPoint), Primavera P6, AutoCAD, Autodesk Inventor, SolidWorks, HTRI, CAESAR II.

Certificate of Competence:

- **Interpretation Drawing P&ID, Isometrics Layout, Rotating, and Static Equipment** [2023]
- **AK3U BNSP** (General Occupational Safety and Health Expert) [2025]
- **Manajer Energi BNSP** (Energy Manager) ISO 50001:2018 [2025]
- **PPPU BNSP** (Emission Pollution Manager) [2026]
- **BST/BSS** (Basic Sea Survival, Basic First Aid, Basic Fire Fighting) [2025]